

Feature Engineering Methodology for the GBT+JEPA Trading System

Kripta Studios

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Abstract

The current feature engineering stack combines two philosophies. The first is explicit domain engineering: option Greek exposure, strike geometry, volatility context, intraday location, and support/resistance descriptors are represented as tabular features. The second is learned representation: XInputJEPA converts a two-hour sequence of state and input variables into latent transition features. The final LightGBM model can then decide which explicit and latent features matter for each ticker and regime.

1 Core Principle: Live-Safe State Estimation

Feature engineering is constrained by the live trading problem. At timestamp t , the model may use only:

$$\mathcal{F}_t = \{\text{market data}_{\leq t}, \text{option chain}_{\leq t}, \text{calendar}_t, \text{model state}_{\leq t}\}.$$

It may not use future returns, future option paths, realized future MFE, realized future drawdown, oracle exit, or future best delta. Those quantities are labels or diagnostics, not input features.

2 Normalization Philosophy

2.1 Robust Normalization for JEPA

XInputJEPA uses robust normalization:

$$\tilde{x}_{i,j} = \text{clip} \left(\frac{x_{i,j} - \text{median}_j}{\text{IQR}_j}, -c, c \right),$$

with clip value $c = 10$. Median and interquartile range are fitted only on the training dates for validation experiments and on all available dates for final production fit. Robust scaling is used because option-derived features have heavy tails and occasional extreme values.

2.2 Median Imputation for GBT

The LightGBM direction model uses median imputation:

$$x_{i,j}^{\text{filled}} = \begin{cases} x_{i,j}, & x_{i,j} \text{ finite,} \\ \text{median}_j, & \text{otherwise.} \end{cases}$$

The fitted medians are stored inside each per-ticker joblib artifact. This keeps live inference deterministic and reproducible.

3 Feature Families

3.1 Greek Exposure Features

The most important engineered features are net Greek exposures:

- `net_gamma`: convexity exposure; often associated with stabilizing or destabilizing dealer hedging depending on sign.
- `net_vanna`: sensitivity of delta exposure to volatility; relevant when implied volatility changes.
- `net_charm`: time decay of delta exposure; especially relevant intraday for 0DTE.
- `net_dgex`: delta-gamma exposure proxy used as a directional/hedging pressure descriptor.
- `net_zomma`: sensitivity of gamma to volatility.
- `net_delta`: aggregate directional option exposure.
- `net_vega` and `net_vomma`: volatility exposure and volatility convexity.

These fields are not used as isolated signals. Their predictive value comes from interaction with price location, volatility regime, and recent changes.

3.2 Weekly Versus 0DTE Exposure

The pipeline includes weekly exposure fields such as `wk_net_gamma`, `wk_net_vanna`, and `wk_net_dgex`. Ratios such as `gamma_0dte_vs_wk` and `vanna_0dte_vs_wk` help distinguish a local 0DTE imbalance from a broader expiry structure. This matters because a same-day imbalance may decay quickly, while weekly exposure can act as a larger magnetic or stabilizing structure.

3.3 Regime Flags

Regime features reduce the burden on the GBT to infer sign states repeatedly. Examples include:

- `gamma_regime`,
- `vanna_bullish`,
- `charm_bullish`,
- `dgex_sticky`,
- `zomma_stabilizing`.

These are compact encodings of nonlinear market-structure assumptions. For example, a distance-to-level feature may be bullish in one gamma regime and mean-reverting in another.

3.4 Strike Geometry

The option chain creates a spatial field around spot. The model encodes normalized distances from current spot to important strikes:

$$d_{\text{level}} = \frac{S_t - K_{\text{level}}}{S_t} \times 10,000.$$

Examples include `dist_to_max_gamma`, `dist_to_min_gamma`, `dist_to_zero_gamma`, `dist_to_max_dgex`, and distances to vega/vomma extrema. These features are important because option pressure is spatial: the same net exposure can matter differently when spot is near a large strike concentration.

3.5 Intraday Price Map

The intraday map includes:

- price relative to initial-balance high and low,
- whether price is above, below, or inside the initial balance,
- initial-balance range percentage,
- distance to prior-day initial-balance levels,
- gap percentage and gap direction,
- minutes to close.

These features encode whether the market is extended, balanced, rejecting, or grinding. They also allow the model to treat morning and afternoon signals differently.

3.6 Volatility Features

Volatility state is represented by:

- `atm_iv`,
- `iv_zscore`,
- `iv_percentile`,
- `vix_spot`,
- `vix_gamma`,
- `vix_regime`,
- `atr_5d_norm`.

These variables condition the model on whether current option prices are rich, cheap, compressed, or expanding. The same directional forecast can have different option payoff properties under different IV states.

3.7 Confluence Features

Confluence features combine level geometry with Greek concentrations. Examples:

- `confluence_ib_high_max_gamma`,
- `confluence_ib_low_min_gamma`,
- `confluence_fib127_bull_max_dgex`,
- `confluence_fib161_bear_min_dgex`.

These are engineered interaction terms. They allow the GBT to split directly on a semantically meaningful condition instead of having to reconstruct it through multiple tree levels.

4 State/Input Split for JEPA

4.1 State Features

State features describe what the system is now: exposure, volatility, location, levels, confluences, and calendar context.

4.2 Input Features

Input features describe recent perturbations:

$$u_t = [\Delta\Gamma_t, \Delta\text{Vanna}_t, \Delta\text{DGEX}_t, \Delta S_t, \text{momentum}_t, \Delta\text{IV}_t, \dots].$$

This is the key JEPA design choice. In the LeWorldModel-style interpretation, the model predicts future state from current latent state and an input. In our market setting, variations in Greek exposure and spot are the input.

5 JEPA Feature Engineering

5.1 Latent Coordinates

The first 28 JEPA features are raw latent coordinates:

$$z_t \in \mathbb{R}^{16}, \quad u_t \in \mathbb{R}^{12}.$$

These coordinates are not individually named economic variables. They are learned factors. Their value is assessed by whether the downstream GBT improves OOS performance when they are present.

5.2 Latent Velocity

Latent velocity is:

$$v_t^z = \|z_t - z_{t-1}\|_2, \quad v_t^u = \|u_t - u_{t-1}\|_2.$$

High velocity indicates that the latent market state or latent input is changing quickly. This can be useful around regime transitions, volatility shocks, or rapid exposure changes.

5.3 Prediction Dispersion

Prediction dispersion measures disagreement among predicted future latent states. Short dispersion compares horizons 1, 3, and 6. Long dispersion compares horizons 12, 24, and 36:

$$D_{\text{short}} = \text{mean}_{i < j, i, j \in \{1, 3, 6\}} \|\hat{z}_{t+i} - \hat{z}_{t+j}\|_2.$$

High dispersion means the model expects future state to vary by horizon; low dispersion means the projected state is more stable.

5.4 Lagged Prediction Error

Lagged prediction errors are computed when a previous forecast matures:

$$e_{t,h} = \|\hat{z}_{t-h,h} - z_t\|_2.$$

This is live-safe because both terms are known at time t . It is a measure of whether the recent latent world model has been surprised.

5.5 Directional Probabilities

The JEPA class head produces:

$$(p_{\text{short}}, p_{\text{hold}}, p_{\text{long}}).$$

The derived features are:

$$\text{confidence} = \max(p_{\text{short}}, p_{\text{long}}),$$

$$\text{direction score} = p_{\text{long}} - p_{\text{short}},$$

$$H(p) = -\sum_i p_i \log(p_i).$$

These features are not final trade decisions. They are inputs to the GBT.

6 Option Candidate Feature Engineering

The option layer adds features at the candidate level:

- one-hot ticker and side,
- jepa180_prob_up, confidence, direction, and edge,
- entry minute, position in day, and minutes to close,
- target delta and actual delta,
- entry premium, raw premium, and spread-adjusted premium,
- premium-to-spot in bps,
- strike distance in points and bps,
- actual IV, theta, gamma,
- theta over premium,
- gamma notional.

The spread adjustment is deliberately conservative. The code increases the effective entry premium using a delta-dependent spread and a gamma-speed adjustment. This penalizes fragile high-gamma entries.

7 Dynamic Exit Feature Engineering

For learned exits, each open trade is represented every 5 minutes by:

- current PnL percentage,
- current return on risk,
- current premium and premium ratio,
- peak PnL percentage,
- drawdown from peak,
- maximum adverse excursion proxy,
- current delta, IV, and gamma,
- spot return and signed spot return,
- hold minutes and minutes remaining.

The target is future best return on risk from the current state:

$$V^*(s_t) = \max_{\tau \in [t, t+T]} \frac{\text{PnL}_\tau}{\text{risk capital}}.$$

This is used to train a continuation-value model. It is not a feature in live inference.

8 Why Many Base Features Can Coexist with JEPA

The user concern that “the GBT already has many features and might ignore JEPA” is correct in principle. That is not a problem; it is the point of the experiment. If the JEPA block adds no marginal signal, the GBT should ignore it. The strict OOS result shows the opposite: `base_jepa` improves trade metrics relative to `base`. Therefore the JEPA features are not merely being ignored.

9 Feature Promotion Criteria

A feature or feature block should be promoted only if:

1. it is live-safe,
2. it is produced identically in training and live inference,
3. it improves strict OOS or walk-forward results,
4. the improvement survives transaction-cost sensitivity,
5. it does not depend on a single ticker or a single lucky month,
6. it is documented in the artifact metadata or feature-name file.

10 Conclusion

The system uses explicit features to encode known market structure and JEPA features to encode latent temporal dynamics. The combined approach is empirically stronger than either component alone. Future feature engineering should prioritize better option-path state variables, liquidity/slippage realism, and robust learned exit targets rather than simply adding more static features to the already broad base set.